

Competition, Separatrices, and Basins

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Definition: Two-species competition model

Let $x(t) \geq 0$ and $y(t) \geq 0$ denote two competing populations. The model

$$\dot{x} = x(3 - x - 2y), \quad \dot{y} = y(2 - x - y)$$

is a **two-species competition model**: each species has self-limited growth and inhibits the growth of the other species.

Notation: Population state space

The state space for the competition model is the closed first quadrant

$$Q = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\}.$$

Recall: Positive invariance

A set is **positively invariant** if every solution starting in it remains in it for all later times for which the solution exists.

Proposition: Nonnegative populations remain nonnegative

The first quadrant Q is positively invariant.

Proof. On $x = 0$, $\dot{x} = 0$, so the y -axis is invariant. On $y = 0$, $\dot{y} = 0$, so the x -axis is invariant. By uniqueness, a trajectory that starts in Q cannot cross either invariant axis.

Question: What are the nullclines and equilibria?

Find every nullcline and equilibrium in Q .

Solution.

$$\dot{x} = 0 \iff x = 0 \text{ or } x + 2y = 3,$$

$$\dot{y} = 0 \iff y = 0 \text{ or } x + y = 2.$$

The intersections in Q are

$$(0, 0), \quad (3, 0), \quad (0, 2), \quad (1, 1).$$

Moreover,

$$\text{sign } \dot{x} = \text{sign}(3 - x - 2y) \quad (x > 0),$$

Question: What are the nullclines and equilibria? (continued)

$$\text{sign } \dot{y} = \text{sign}(2 - x - y) \quad (y > 0).$$

Visualization: Nullcline sign regions

In the open first quadrant,

inequalities	$(\text{sign } \dot{x}, \text{sign } \dot{y})$	direction
$x + 2y < 3, x + y < 2$	$(+, +)$	
$x + 2y > 3, x + y < 2$	$(-, +)$	
$x + 2y < 3, x + y > 2$	$(+, -)$	
$x + 2y > 3, x + y > 2$	$(-, -)$	

The oblique nullclines have intercepts

$$x + 2y = 3 : (3, 0), (0, 3/2),$$

$$x + y = 2 : (2, 0), (0, 2),$$

and cross at $(1, 1)$.

Question: How is each equilibrium classified?

Compute the Jacobian, eigenvalues, and eigendirections at all four equilibria.

Solution.

$$J(x, y) = \begin{pmatrix} 3 - 2x - 2y & -2x \\ -y & 2 - x - 2y \end{pmatrix}.$$

At $(0, 0)$,

$$J(0, 0) = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix},$$

so the equilibrium is an unstable node.

At $(0, 2)$,

$$J(0, 2) = \begin{pmatrix} -1 & 0 \\ -2 & -2 \end{pmatrix},$$

with eigenpairs

$$\lambda = -1, \mathbf{v} = (1, -2), \quad \lambda = -2, \mathbf{v} = (0, 1).$$

Thus $(0, 2)$ is an asymptotically stable node.

At $(3, 0)$,

$$J(3, 0) = \begin{pmatrix} -3 & -6 \\ 0 & -1 \end{pmatrix},$$

with eigenpairs

$$\lambda = -1, \mathbf{v} = (3, -1), \quad \lambda = -3, \mathbf{v} = (1, 0).$$

Thus $(3, 0)$ is an asymptotically stable node.

At $(1, 1)$,

$$J(1, 1) = \begin{pmatrix} -1 & -2 \\ -1 & -1 \end{pmatrix},$$

Question: How is each equilibrium classified? (continued)

with eigenpairs

$$\begin{aligned}\lambda_s &= -1 - \sqrt{2}, & \mathbf{v}_s &= (\sqrt{2}, 1), \\ \lambda_u &= -1 + \sqrt{2}, & \mathbf{v}_u &= (\sqrt{2}, -1).\end{aligned}$$

Thus $(1, 1)$ is a saddle. Its stable direction has slope $1/\sqrt{2}$ and its unstable direction has slope $-1/\sqrt{2}$.

Theorem: Local invariant manifolds of a hyperbolic saddle

Let $U \subseteq \mathbb{R}^2$ be open, let $\mathbf{F} : U \rightarrow \mathbb{R}^2$ be continuously differentiable, and let $p \in U$ be a hyperbolic saddle. There is a neighborhood $N \subseteq U$ of p and unique continuously differentiable curves

$$W_{\text{loc}}^s(p), \quad W_{\text{loc}}^u(p)$$

through p , tangent at p to the stable and unstable eigenspaces of $D\mathbf{F}(p)$, respectively. They are characterized by

$$W_{\text{loc}}^s(p) = \{\mathbf{z}_0 \in N : \Phi_t(\mathbf{z}_0) \text{ is defined and belongs to } N \text{ for every } t \geq 0, \Phi_t(\mathbf{z}_0) \rightarrow p \text{ as } t \rightarrow \infty\},$$

$$W_{\text{loc}}^u(p) = \{\mathbf{z}_0 \in N : \Phi_t(\mathbf{z}_0) \text{ is defined and belongs to } N \text{ for every } t \leq 0, \Phi_t(\mathbf{z}_0) \rightarrow p \text{ as } t \rightarrow -\infty\}.$$

Recall: Basin of attraction

Let p be an attracting equilibrium in a state space D . Its basin of **attraction** relative to D is

$$\mathcal{B}_D(p) = \left\{ \mathbf{z}_0 \in D : \begin{array}{l} \Phi_t(\mathbf{z}_0) \text{ exists for every } t \geq 0, \\ \Phi_t(\mathbf{z}_0) \rightarrow p \text{ as } t \rightarrow \infty \end{array} \right\}.$$

Definition: Relative closure

Let $A \subseteq D \subseteq \mathbb{R}^2$. The **closure of A relative to D** is

$$\overline{A}^D = \left\{ p \in D : \begin{array}{l} \text{for every } \varepsilon > 0 \text{ there is a } q \in A \\ \text{such that } \|q - p\| < \varepsilon \end{array} \right\}.$$

Definition: Relative boundary

Let $A \subseteq D \subseteq \mathbb{R}^2$. The **boundary of A relative to D** is

$$\partial_D A = \overline{A}^D \cap \overline{D \setminus A}^D,$$

where both closures are taken relative to D .

Definition: Basin boundary

Let $\mathcal{B}_D(p)$ be a basin of attraction in a state space D . Its **basin boundary** is its relative topological boundary

$$\partial_D \mathcal{B}_D(p).$$

Definition: Basin separatrix

Let p and q be distinct attracting equilibria in D . An invariant curve $S \subseteq D$ contained in

$$\partial_D \mathcal{B}_D(p) \cap \partial_D \mathcal{B}_D(q)$$

is a **basin separatrix** for p and q .

Proposition: A compact forward-invariant rectangle

Let $(x_0, y_0) \in Q$ and define

$$M_x = \max\{x_0, 3\}, \quad M_y = \max\{y_0, 2\}.$$

Then

$$R = [0, M_x] \times [0, M_y]$$

is positively invariant. Consequently, the solution exists for all $t \geq 0$ and remains bounded.

Proof. On the right edge $x = M_x$,

$$\dot{x} = M_x(3 - M_x - 2y) \leq 0.$$

On the top edge $y = M_y$,

$$\dot{y} = M_y(2 - x - M_y) \leq 0.$$

The coordinate axes are invariant, so the vector field never points out through the remaining edges. Hence R is positively invariant. The vector field is smooth and the trajectory remains in the compact set R , so the continuation theorem gives existence for all forward time.

Proposition: No periodic orbit lies in the open first quadrant

The competition model has no periodic trajectory contained in

$$Q^\circ = \{(x, y) : x > 0, y > 0\}.$$

Proof. Define

$$B(x, y) = \frac{1}{xy}$$

on Q° . If $\mathbf{F}$ is the competition vector field, then

$$B\mathbf{F} = \left(\frac{3 - x - 2y}{y}, \frac{2 - x - y}{x} \right)$$

and

$$\operatorname{div}(B\mathbf{F}) = -\frac{1}{y} - \frac{1}{x} < 0.$$

Proposition: No periodic orbit lies in the open first quadrant (continued)

Suppose a periodic trajectory $C \subset Q^\circ$ encloses a region U . The vector field $B\mathbf{F}$ is tangent to C , so its outward flux across C is zero. If $\mathbf{n}$ denotes the outward unit normal, Green's theorem gives

$$0 = \int_C B\mathbf{F} \cdot \mathbf{n} \, ds = \iint_U \operatorname{div}(B\mathbf{F}) \, dA < 0,$$

a contradiction.

Observation: What the trapping calculations do not prove

The compact forward-invariant rectangle and the exclusion of periodic orbits do not, by themselves, prove that every solution converges to an equilibrium or determine the global shape of the saddle's stable manifold.

Visualization: Schematic separatrix and basin portrait

Let

$$\Sigma = W^s((1, 1)) \cap Q, \quad S = \bar{\Sigma}^Q.$$

Locally at $(1, 1)$, Σ is tangent to $(\sqrt{2}, 1)$ and therefore has positive slope. The schematic global portrait has the organization

schematic initial position	schematic forward limit
component of $Q \setminus S$ containing $(3, 0)$	$(3, 0)$
on Σ	$(1, 1)$
$(0, 0)$	$(0, 0)$
component of $Q \setminus S$ containing $(0, 2)$	$(0, 2)$

In this schematic, Σ is drawn as the invariant basin separatrix and S as the common relative boundary of the two attracting basins. Establishing those global boundary claims requires arguments beyond the local, forward-invariance, and no-cycle calculations above.

Observation: Dynamics on the coordinate axes

The axes contain the logistic flows

$$y = 0 : \dot{x} = x(3 - x), \quad x = 0 : \dot{y} = y(2 - y).$$

Question: Does one local velocity determine the surviving species?

Compute the vector field at $(x, y) = (1/2, 1/2)$.

Solution.

$$\begin{aligned} \dot{x} &= \frac{1}{2} \left(3 - \frac{1}{2} - 1 \right) = \frac{3}{4}, \\ \dot{y} &= \frac{1}{2} \left(2 - \frac{1}{2} - \frac{1}{2} \right) = \frac{1}{2}. \end{aligned}$$

The trajectory initially moves northeast. This one velocity does not identify its forward limit; that requires locating the initial state relative to the global stable curve $\Sigma = W^s((1, 1)) \cap Q$ shown in the schematic portrait.

Visualization: Local data and global data

local data	global data
$J(p)$	$\mathcal{B}_Q(p)$
eigenvalues and eigendirections	$W^s((1, 1))$
nullcline crossing direction	separatrix location

Observation: Local classification does not locate a basin boundary

The Jacobian proves that $(3, 0)$ and $(0, 2)$ attract nearby states. Deciding which sink attracts a distant state requires global information, including the location of the saddle's stable manifold.